

```
a leap year";
    else
        cout<<"You have entered "<<year<<" and this is
not a leap year";
    }
    else
    {
        if(year%4==0)
            cout<<"You have entered "<<year<<" and this is
a leap year";
        else
            cout<<"You have entered "<<year<<" and this is
not a leap year";
    }
}
```

The output of this program is as follows:

```
Please enter any year of your choice => 1998
You have entered 1998 and this is not a leap year
```

Before we proceed with the explanation of the program, we need to know the criterion that decides a leap year. If a year is divisible by 400, then it is a leap year.

In the above program, the year i.e. the integer value is accepted and stored in an integer variable. The 'if' statement checks whether the value is divisible by 100. If the condition returns true, then the second 'if' statement checks whether the value is divisible by 400. If this condition is also true, then the given value i.e. the year will be a leap year and a message is displayed with the value entered by the user. If the second 'if' statement returns false, then the year will not be a leap year and a corresponding message is displayed along with the value of the year to show the result. Now if the first 'if' statement returns false, i.e. if the value is not divisible by 100, then the control moves to the 'else' part that checks whether the value is divisible by 400 or not by the second 'if' statement. If this holds true, then the